Main Report — HW03: GUI and Usability Testing

Field	Value
Student	Nguyễn Thành Dâng
Student ID	23127334
Identity overlay	`23127334@hcmus.edu.vn`
SUT	EShop (`eshop-sut`) — Vietnamese e-commerce demo
Repository under test	https://github.com/ttbhanh/eshop-sut
GitHub Issues (bugs)	https://github.com/ThanhDang-Vn/software-testing/issues (#18–#29)
Usability scale	SUS (System Usability Scale)
AI policy	Open — AI used and fully declared (see `reports/ai-audit-report.md` , `reports/ai-critique.md`)

This report documents the full process for the three required tasks — GUI checklist design and execution, a moderated 7-participant usability evaluation, and cross-browser / cross-platform testing — together with the agent skill and the mandatory AI appendices. Every AI-produced artifact in this package was reviewed against the running SUT and the actual requirements before it was kept.

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1. Scope, environment, and selection

1.1 Selected screens and flow

Following §5 (Scope Selection), the work targets the SUT's **user interface**, concentrating on two screens joined by one end-to-end flow so that the checklist and the usability study reinforce each other.

Scope	Route	Primary requirement	Role in this HW
Login	`/login`	FR-02 (Login & account lockout)	GUI checklist + entry point of the usability flow
Order History (inside Profile)	`/profile`	FR-11 (Order history view, user)	GUI checklist + goal screen of the usability flow

Selected usability flow (single end-to-end flow): *sign in with a supplied account → independently locate the personal order history → identify the latest order's ID, date, total, and current status.* This same flow is the task scenario used with the seven participants in Task 2 and the flow exercised on every platform in Task 3, giving the three tasks a shared, comparable baseline.

The student confirms this primary screen and flow are **not duplicated** by any other member of the group (per §5).

1.2 Environment

Item	Value
Frontend	`http://localhost:5173` (Vite)
Backend API	`http://localhost:3000`
Checklist execution	Playwright 1.55.0, Chromium, Windows
Main viewport	1280×800; plus 320×640, 200% zoom, portrait/landscape, RTL, dark preference
Cross-platform	BrowserStack Live (trial); local SUT exposed through Cloudflare quick tunnels
Test data	Five orders seeded through authenticated APIs so all five order-status labels are observable

Results were observed **at runtime**. Source inspection (`Login.jsx` , `Profile.jsx` , `App.jsx` , `AuthContext.jsx` , and the backend login/order endpoints) was used only to *design* checks and to explain observed behaviour — never to infer a pass/fail without running the SUT.

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2. Interface analysis (basis for the checklist)

2.1 User goals

- Sign in safely and understand clearly whether authentication succeeded or failed.
- Recover from a typing or credential error without exposing account details.
- Locate personal order history after authentication, unaided.
- Identify the newest order and correctly read its ID, date, total, and state.

2.2 Main UI components and forms

- **Login** — email/username field, password field, "Quên mật khẩu?" link, submit button, registration link, and credential-error feedback.
- **Order History** — shares the Profile page with the profile-edit form; renders either an empty message or a five-column table (Mã ĐH, Ngày đặt, Tổng tiền, Trạng thái, Thao tác) with a per-row cancel action for eligible orders.

2.3 Navigation paths

`Header → Đăng nhập → /login → successful login → Home → account greeting → /profile → Lịch sử đơn hàng`. Direct-URL access and expired-session behaviour are also considered. A key observation feeding both tasks: **there is no explicitly named "Order History" navigation entry** — the only route into history is the account greeting, which does not read as a link.

2.4 Feedback and state

Important states enumerated for the checklist: initial, invalid native form input, wrong credentials, temporary lockout, successful login, order loading, empty history, populated history, API failure, and session expiration.

2.5 Interface risks (the four IAs)

- **IA-01 General UI standards** — contrast, target size, responsive reflow, zoom, text spacing, orientation, colour-independent status, localization of date/VND.
- **IA-02 Forms** — input semantics (`type=email` / `password`), label association, required markers, autocomplete, keyboard order, focus visibility, error linkage.
- **IA-03 Navigation** — discoverability of order history and current-location indication.
- **IA-04 Feedback / state** — pending state on submit, loading indicator, distinct error-vs-empty state, cancel confirmation and result feedback, live announcement of dynamic errors / status changes.

Exploratory risks explicitly flagged (no explicit SUT requirement): **RTL layout** and **dark mode**.

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3. Task 1 — GUI checklist

3.1 Design method (AI-first, then human review)

The checklist was built in the disciplined, staged way §2 requires rather than by a single generic prompt:

1. **AI baseline (`gui-checklist-v0-agent.xlsx`, 32 items).** An AI tool generated an initial checklist across the two screens.
2. **Standards-based human review (`gui-checklist-v1-reviewed.xlsx`, 45 items).** Every AI item was re-derived against **WCAG 2.2, WAI Forms, ARIA APG, the GOV.UK Design System, and Nielsen's heuristics**. Items the AI missed were added, each with a written reason; items that mixed a GUI oracle with a non-GUI (auth/API/session/security) assertion were split so each case has **exactly one observable objective and one oracle**.

The full V0→V1 comparison, the reason the AI missed each added item, the scope-migration of non-GUI items, and the lessons learned are in `task-1-gui-checklist/human-review.md`. Both workbooks are submitted so the raw AI baseline stays visible as evidence; no `Source` column is kept in the delivered workbooks (the provenance is documented separately in `human-review.md`).

What the AI systematically missed (representative, not exhaustive): password manager / autocomplete purpose, 200% zoom reflow on the data table, RTL, submit pending-state, colour-independent status, table header–cell association and `<caption>`, screen-reader live-region announcement of dynamic errors and status changes, WCAG *Focus Not Obscured*, text-spacing overrides, orientation, and dark mode. The recurring cause: a generic "generate a GUI checklist" prompt does not specify the WCAG level, the assistive tooling, the viewports, or the one-objective-per-case rule, so the model defaulted to happy-path, by-eye checks.

3.2 Coverage

45 atomic items (> 40 minimum) spanning both screens and **all four IAs**:

Interface aspect	Example item IDs	Covered
IA-01 General UI standards	GUI-L-001/003/004/005/006/021/028/029/030, GUI-O-002A/004/005/006/007/008/018/019/020/021/025/028	✅
IA-02 Forms	GUI-L-007/008/009/010/011A/013/014/020/025/027, GUI-O-010/011/027	✅
IA-03 Navigation	GUI-O-012, GUI-O-013	✅
IA-04 Feedback / state	GUI-L-024A/026, GUI-O-016/022/024A/029	✅

3.3 Execution

The 45-item V1 checklist was executed with Playwright 1.55.0 on Chromium against the running local frontend and backend. Each item carries an **Expected result, Actual result, Status, Notes, Evidence, and a consolidated Bug ID**.

Result	Count
Passed	17
Failed	28
Blocked	0
Total	**45**

Every **Failed** item has an actual result, a note explaining *why* it failed, and a screenshot under `task-1-gui-checklist/failed-screenshots/` (28 screenshots, Failed items only, per §6). Two items that depend on assistive technology (GUI-O-026 reading order, GUI-O-029 status announcement) were not inferred from source code: they were executed with a **manual screen-reader run** and both **passed**, so all

45 items now carry a Pass/Fail verdict with no items left Blocked. A per-screen and per-IA breakdown is in `task-1-gui-checklist/test-summary.md`.

3.4 Bugs (12), reported in Markdown and on GitHub Issues

The 28 Failed items were consolidated by root cause into **12 GUI bugs**, each reported both in `task-1-gui-checklist/bug-report.md` and as a GitHub issue (#18–#29, label `homework3`) with a screenshot attached.

Bug	Screen	Summary	Severity	Issue
BUG-GUI-001	Login	No `h1`; heading reads "Đăng Ký"; "Username"/"Sign In" mix EN with VI	Major	#18
BUG-GUI-002	Login	Blue link below 4.5:1 contrast; several targets under 44×44 px	Major	#19
BUG-GUI-003	Login	Email/password use `type=text` (password visible); no autocomplete tokens	Critical	#20
BUG-GUI-004	Login	No required markers, no label `for/id`, no field-level or live error semantics	Major	#21
BUG-GUI-005	Login	Positive `tabindex` breaks keyboard focus order	Major	#22
BUG-GUI-006	Login	Submit shows no pending state; repeat activation not prevented	Major	#23
BUG-GUI-007	Order History	Non-Vietnamese currency grouping and US date order (`7/26/2026`)	Minor	#24
BUG-GUI-008	Order History	Cancel executes with no confirmation dialog / focus management	Critical	#25
BUG-GUI-009	Order History	Order history not discoverable; no current-location indicator	Major	#26
BUG-GUI-010	Order History	No loading state; API failure indistinguishable from empty history	Major	#27
BUG-GUI-011	Order History	Table has column headers but no `<caption>` / accessible name	Major	#28
BUG-GUI-012	Order History	Five-column table causes page-wide horizontal overflow at 320 px / 200%	Major	#29

4. Task 2 — Usability evaluation (7 participants)

4.1 Plan and objectives

Objective (`objective.md`): determine whether typical Vietnamese online shoppers can sign in, *independently* discover their order history, and correctly interpret the latest order's four FR-11 fields without moderator guidance. Research questions cover login error recovery, discoverability, correct field reporting, label/colour comprehension, and trust.

Target user profile: 18+, has used e-commerce at least once, **not enrolled in this HW03 class**, preferably non-IT / non-tester, reads Vietnamese, consents.

Task scenario (`task-scenario.md`) — goal, not steps: "*Bạn đã từng mua hàng trên EShop và muốn kiểm tra lại một đơn hàng gần đây. Hãy đăng nhập bằng tài khoản được cung cấp, tìm nơi chứa các đơn hàng trước đây, rồi cho biết mã đơn, ngày đặt, tổng tiền và trạng thái hiện tại của đơn hàng gần nhất.*" The words `/profile`, "Lịch sử đơn hàng", and any step-by-step navigation are deliberately withheld from the participant.

Instruments: the 10-item **SUS** completed after every session (`sus-questionnaire.md`), plus open probe questions covering, at minimum, **clarity, error recovery, speed, and trust** (`sessions/P0x/probe-answers.md`).

Completion coding: `SUCCESS_UNASSISTED`, `SUCCESS_ASSISTED` (after a necessary neutral intervention), or `FAIL` (cannot report all four fields within 8 minutes).

Pilot: a moderator dry-run was completed and marked READY before the official sessions (`pilot-session/pilot-notes.md`), confirming the scenario wording, the account state, and timing.

4.2 Conduct

Seven real participants (P01–P07), recruited **outside the HW03 class**, each completed one moderated session. The moderator set the stage ("we test the product, not you"; think-aloud), observed neutrally, stepped in only when a participant was fully stuck, and closed with the SUS scale and probe questions. Every session was **screen-recorded with consent** (unlisted YouTube links in `recording-links.md`; consent tracked in `consent-statement.md`). A verifiable **email contact** for each of the seven participants is listed in `participant-list.csv` / `.xlsx` for TA verification (per §11).

4.3 Results (n = 7)

Metric	Value
Completion	7/7 (100%)
Unassisted completion	5/7 (P01, P03, P04, P05, P07)
Assisted completion	2/7 (P02, P06)

Metric	Value
Mean completion time	287.3 s (4:47)
Mean SUS	**66.4 / 100**
Errors / wrong turns / hesitations / interventions	1 / 4 / 19 / 2

Per-participant SUS (`sus-summary.csv`):

P01	P02	P03	P04	P05	P06	P07	Mean
80.0	57.5	75.0	55.0	87.5	32.5	77.5	**66.4**

Mean SUS 66.4 sits just below the ~68 acceptability benchmark, which is consistent with the discoverability friction observed on the flow.

4.4 Analysis — severity-ranked findings

Notes were synthesised, similar pain points grouped, and isolated bugs separated from systemic design issues (`severity-ranked-findings.md`):

Rank	ID	Severity	Freq	Finding	Recommendation
1	USAB-01	Critical	7/7	Order-history entry point not discoverable; account greeting not read as a link	Add an explicit "Đơn hàng của tôi" nav entry; mark current page
2	USAB-07	Major	5/7	History lacks per-order product detail and actions; no cancel confirmation	Order detail view; confirm-before-cancel; back-to-shop link
3	USAB-04	Major	4/7	Password not masked on login; EN/VI label mix	Mask password (with show/hide); consistent Vietnamese labels
4	USAB-03	Major	4/7	Ambiguous US date format; no time shown	Deterministic `31/07/2026`, `450.000 ₫`, include time
5	USAB-06	Major	3/7	Table sort order not labelled	Label "Mới nhất trước"; sortable "Ngày đặt" header
6	USAB-05	Minor	1/7	Unexpected keyboard focus order on login	Remove positive `tabindex`
7	USAB-08	Minor	2/7	Red total reads as a warning; prominent exit; order code felt unnecessary	Neutral colour for totals; de-emphasise exit

The dominant, **critical** finding (USAB-01) is the same discoverability defect captured by BUG-GUI-009 in Task 1 and confirmed cross-platform in Task 3 — triangulated evidence from three independent methods. The primary recommendation is a clearly labelled order-history navigation entry, plus deterministic Vietnamese date/currency formatting and a masked password field.

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5. Task 3 — Cross-browser / cross-platform

The Task-2 flow (**Login → Hồ sơ → Lịch sử đơn hàng, FR-11**) was run on **BrowserStack Live**, with the local SUT exposed through two Cloudflare quick tunnels (frontend + backend). Seven screenshots were captured; each shows the OS/browser/device in the BrowserStack toolbar, the SUT tunnel URL, and the student identity `23127334@hcmus.edu.vn` (typed into Username at login; shown as the Profile email and the greeting "Chào, Nguyễn Thành Dâng" plus the "Họ Tên" field on the Order-History screens).

Platform	OS / device	Screens	Screenshots	Result
Google Chrome 147	macOS desktop	Login; Order History	`chrome1.png`; `chrome2.png`	Rendered correctly
Mozilla Firefox 144–145	Windows 11 desktop	Login; Order History	`firefox1.png`; `firefox2.png`	Rendered correctly
Safari (iOS)	iPhone 17 / iOS 26.4; iPhone 16 / iOS 27.0	Login; Order History	`safari1.png`; `safari2.png`; `safari3.png`	Rendered correctly (mobile reflow)

This is a **genuine Safari-on-iOS** run — it satisfies the strict "Chrome, Firefox, and Safari" requirement without a WebKit-on-Windows substitute. Fifteen atomic cross-platform checks (CP-T01–CP-T15) returned **14 Passed / 1 Failed**; the single failure (CP-T15) is the order-table horizontal overflow on the narrow mobile viewport — i.e. the Task-1 responsive defect (BUG-GUI-012), confirmed as a **consistent application-level** issue rather than a one-platform break. The US-style date (`7/31/2026`) and comma-grouped currency (`62,000,000 đ`) render **identically on every platform**, confirming the locale defects (BUG-GUI-007) are application-level, not browser-specific. Full detail is in `task-3-cross-platform/test-summary.md`.

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6. Agent skill

An agent skill (`agent-skills/hw03-gui-usability-testing.skill.md`) encodes the GUI-checklist and usability-evaluation workflow — required inputs, mandatory gates (>40 items, all four IAs, source/expected/actual/status/notes per item, screenshots for failed items only), and the usability protocol — so it can be reused on additional screens and flows. Demonstration video link(s) are recorded in `agent-skills/demo-video-links.md`.

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7. AI usage (summary)

AI was used and is fully declared. The complete, per-interaction log (tool, date, prompt, output, human review) is in `reports/ai-audit-report.md`; the mandatory 200–300-word critique is in `reports/ai-critique.md`. In short: AI accelerated scaffolding (checklist baseline, execution harness, document drafting, tunnel setup) but repeatedly produced work that *looked* finished yet was wrong until checked against the running SUT and the real requirements — most notably an initial batch of *simulated* usability personas that had to be replaced with genuine recorded sessions, and a WebKit-on-Windows run that had to be redone as real Safari-on-iOS. Every such artifact was corrected or regenerated under human review before inclusion.

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8. Limitations

- The two assistive-technology items (GUI-O-026 reading order, GUI-O-029 live status announcement) were verified with a single manual screen-reader run; broader AT coverage across multiple screen readers was not performed.
- All usability sessions ran on **laptop/desktop**; mobile responsiveness was not evaluated with a real handset in a moderated session (Task 3 did exercise mobile Safari, but not with a participant).
- The sample (n = 7) follows the "discount usability" design: sufficient to expose repeated problems, not intended for quantitative generalisation.
- Participants are HCMUS students; recruitment ensured **none are enrolled in HW03** (a hard eligibility rule); non-IT / non-tester is a preference only.

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9. Deliverables index

Deliverable	Location
Main report (MD + PDF)	`reports/main-report.md`, `reports/main-report.pdf`
GUI checklist (Excel v0 + v1) + review	`task-1-gui-checklist/gui-checklist-v0-agent.xlsx`, `gui-checklist-v1-reviewed.xlsx`, `human-review.md`
Checklist execution + bugs	`task-1-gui-checklist/execution-results.json`, `bug-report.md`, `failed-screenshots/`, `github-issues-links.md`

Deliverable	Location
Usability plan + sessions	`task-2-usability/` (objective, task-scenario, sus-questionnaire, moderator-script, consent-statement, pilot-session, sessions/P01–P07)
SUS + findings + participants	`task-2-usability/sus-summary.csv` , `sus-ueqs-summary.xlsx` , `severity-ranked-findings.md` , `participant-list.csv/.xlsx` , `recording-links.md`
Cross-platform	`task-3-cross-platform/test-summary.md` , `screenshots/`
Agent skill + demo	`agent-skills/hw03-gui-usability-testing.skill.md` , `demo-video-links.md`
AI appendices (MD + PDF)	`reports/ai-audit-report.md/.pdf` , `reports/ai-critique.md/.pdf`
Git commit log	`git-commit-log.txt`
README (self-assessment + summary)	`README.md`